You want your cable drum motor to rotate automatically in both CW (clockwise) and ACW/CCW (anticlockwise) directions, so you don’t have to change the polarity manually.

For this, we’ll build an automatic motor reversing system.

However, because your motor is a BLDC motor, there is an important point: a standard brushed DC motor H-bridge is not used with a BLDC motor. A BLDC controller / ESC is used instead.

1. How Does an Automatic Reversing System Work?

The operating sequence for your cable drum would be:

1. The controller receives a START command.

2. The motor runs CW and winds cable onto the drum.

3. When the specified cable length or end position is reached, a sensor sends a signal.

4. The controller brings the motor speed to zero.

5. The motor then runs CCW if that operation is required.

6. The motor stops when the second sensor is triggered.

Sensors and programmed logic automate this sequence.

2. What Is an H-Bridge?

An H-bridge is an electronic circuit containing four switches, such as MOSFETs or transistors.

It changes the voltage polarity at a DC motor’s terminals.

Switch operation Motor direction

Q1 + Q4 ON CW

Q2 + Q3 ON CCW

All OFF Stop (coast, unless a braking circuit is used)

Note: The two MOSFETs on the same leg must not be ON at the same time. This can cause a short circuit, called shoot-through.

3. What Is Used with Your BLDC Motor?

Your BLDC motor system would use these components:

Component Function

22.2 V battery DC power supply

Bidirectional BLDC controller / ESC Controls motor speed and direction

ESP32 Automatic control and programming

Limit switch / sensor Detects the end position

Emergency stop Shuts down the system in an emergency

BLDC motor Rotates the drum CW or CCW

A BLDC motor has three phase wires: U, V, and W. The ESC switches these electronically to turn the motor. A reversible BLDC ESC can run the motor in both directions when given the appropriate direction command.

4. Automatic Control Block Diagram

22.2 V battery

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BLDC controller / ESC

↓

BLDC motor

↓

Flexible coupling + drum shaft

ESP32 controller

↓

Sends direction and speed commands to the ESC

Sensors + limit switches + emergency stop

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Control signals and feedback

5. Recommendation for Your Project

Recommended control architecture

- 22.2 V BLDC motor
- Bidirectional BLDC controller
- ESP32
- Encoder to measure drum RPM and rotation
- Cable tension sensor
- Limit switches
- Emergency stop

The motor must be brought to a controlled stop before reversing. The system also needs suitable braking and over-tension protection to prevent the drum from reversing abruptly.

One important question

What type of motor do you have?

1. Motor type
   - BLDC motor (three motor wires)
   - Brushed DC motor (two motor wires)
2. How would you like automatic reversing to work?
   - Change direction automatically using a sensor
   - Change direction after a set time
   - Control direction using a switch or remote

I can use this information to explain the appropriate circuit connections and wiring diagram.

My Motor Is BLDC, and I Want Automatic Control. Explain the Automatic Reversing Circuit for My 22.2 V Cable Drum Project.

For your 22.2 V BLDC motor and 50 kg fibre-optic cable drum, we can use an ESP32-based automatic forward–reverse control system.

The motor will automatically rotate CW and CCW. You will not need to reverse the battery polarity manually.

The most important point is that we will not use a standard H-bridge with a BLDC motor. We will use a bidirectional BLDC controller (ESC), which controls the motor's three phases electronically.

1. Complete System Architecture

22.2 V LiPo battery (6S)

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Main fuse + main contactor + emergency stop

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Bidirectional BLDC controller

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BLDC motor → flexible coupling → drum shaft

ESP32 microcontroller

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Automatic control logic

↓

Control signals and feedback

Encoder + tension sensor + limit switches

2. Components and Their Connections

Component	Function
22.2 V LiPo battery (6S)	Main power supply
Bidirectional BLDC ESC (VESC-type)	Controls CW/CCW rotation and speed
ESP32	Runs the automatic operation
Rotary encoder	Measures drum RPM and rotation
S-type load cell + HX711	Measures cable tension
Forward/reverse limit switches	Detect end positions
Emergency stop	Disables the system in an emergency
Main fuse	Protects against electrical short circuits

Controller selection requirement: The ESC must support a 6S battery. A fully charged 6S battery reaches 25.2 V, so do not choose an ESC based only on the battery’s 22.2 V nominal rating.

A VESC-type controller can change direction using positive and negative RPM commands. Its documentation may also provide UART/CAN control and reverse RPM commands.